
O-RAN Open X-haul Transport Working Group

Architecture and Requirements for RAN Fronthaul over Converged Packet Transport Network

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Foreword

This Technical Specification (TS) has been produced by WG9 of the O-RAN ALLIANCE.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN ALLIANCE modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

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Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

Executive summary

This document provides a functional reference architecture of the RAN Fronthaul over Converged Packet Transport Networks. It is intended to be used by operators to specify the building blocks of their fronthaul networks and to provide some deployment guidance. It is intended to be used by vendors to select what to implement and support in their products.

Introduction

This specification is an implementation agreement. It does not define technology, but instead references standardized technologies from other SDOs, e.g., ITU-T, IEEE, IETF, to establish requirements that ensure the realization of a packet transport network for RAN Fronthaul.

Several fundamental changes in the radio access network (RAN) architecture were introduced in the evolution from 4G to 5G [3]. Utilizing new spectrum and beamforming radios results in a significant increase in radio capacity for 5G. The novel use cases asked for more deployment flexibility. These changes triggered the introduction of new splits in the RAN protocol stack. Figure 1 shows an example of the lower layer split (LLS), also known as fronthaul. Fronthaul, in conjunction with midhaul and backhaul, creates the end-to-end 5G RAN transport.

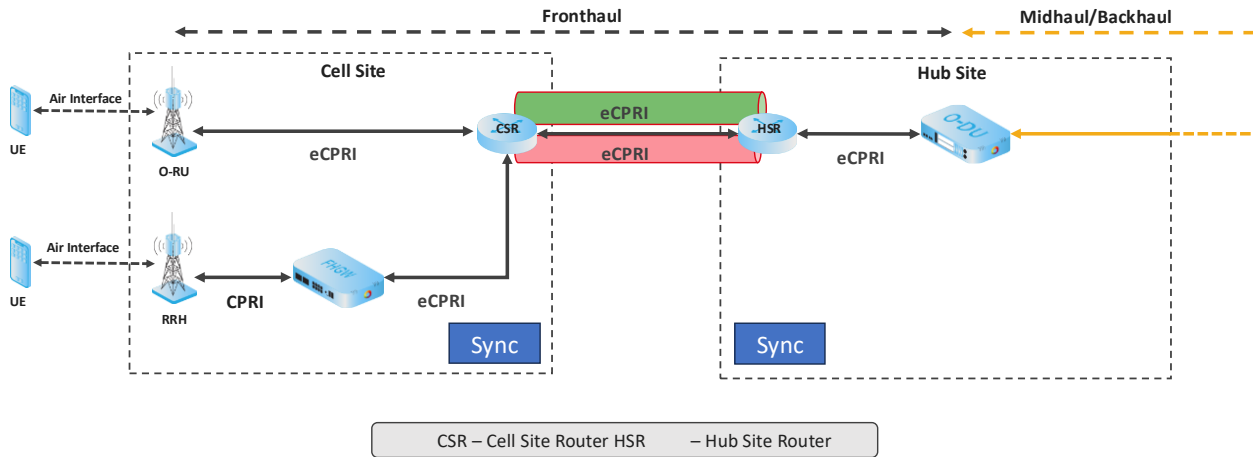


Figure 1 An example of RAN Fronthaul

The 5G architecture and service expectations, including ultra-reliable high bandwidth, low latency, and jitter communication, demand a resilient and scalable network. With the introduction of the eCPRI [4] interface, a packetized Ethernet-based transport could be utilized in the fronthaul. In addition to Ethernet, a packetized fronthaul can leverage IP/MPLS transport technology to provide highly resilient and traffic-engineered transport. Most current dynamic IP routing protocols can provide a highly scalable and resilient underlay network for this architecture. In addition, the Service Level Agreement (SLA) of each application can be guaranteed via a service-aware transport network, assigning a service slice for the appropriate application based on its throughput and latency needs, among other criteria.

The transport itself can be built from existing technologies like segment routing and SRv6 to provide the traffic engineering, resiliency, and Operations, Administration, and Maintenance (OAM) tools needed for fronthaul. Services like EVPN or IP-VPN can be created over this transport to provide transport with an appropriate SLA for Layer 2 eCPRI and, in the future, Layer 3 eCPRI. This packetized fronthaul can provide transportation for other applications as well, which is beyond the scope of this document.

This document outlines the necessary transport service-specific requirements for the fronthaul and coordination of the interface to comply with the increased demands of 5G and beyond 5G networks. It lists the required functions to provide flexibility in mixing and matching different deployment scenarios, functional splits, and potentially other services on the same packet network.

1 Scope

Fronthaul networks have to provide flexibility to mix and match different deployment scenarios (CRAN, DRAN), functional splits, and other mobile traffic on the same packet network. CPRI remains the preferred choice for legacy (2G/3G) support. For 4G/5G technologies and beyond, the novel LLS and eCPRI together are capable of reducing bitrate requirements on the fronthaul for classic radios, preventing the rate explosion for massive MIMO, and enabling connectivity flexibility in packet fronthaul.

The majority of today's deployments are DRANs, typically using wireless microwave links or leased lines transporting packets for S1/NG and X2 backhaul. However, CRANs enable key potential benefits, such as simplification antenna sites, full inter-site coordination, and operational gains by centralizing processing resources at hub sites. The flexible relationship between radios and physical ports on a baseband together with the user data-dependent characteristics of eCPRI/LLS, allows a baseband node to serve a larger set of radios and utilize processing capacity more efficiently.

Several important characteristics of packet-based fronthaul networks need to be met, such as

- Efficiency – make use of the statistical aggregation possibilities in eCPRI/LLS,
- Flexibility – allow for connectivity between all radios and all basebands,
- Scalability – small to large CRAN deployments using the same architectural principles,
- Resiliency – a robust network to minimize the effects of failures,
- Observability – based on the telemetry.
- Security – via service isolation, integrity and privacy protection.
- Simplicity – a network easy to dimension and deploy,
- RAN-agnostic – support of both purpose-built and cloud-native RAN networks,
- Synchronization – providing sync for the RAN components.

A service overlay network architecture enables the separation of the physical network topology from the application service structure and connectivity. Packet-based transport can utilize protocols that have been broadly deployed and well-understood for years for sharing network topology and characteristic information. For example, the eCPRI L2 Ethernet traffic can be handled as a virtual overlay Ethernet service with fronthaul-specific strict service requirements. Such architecture can combine additional virtual overlay Ethernet or IP services with tailored characteristics.

This networking concept provides ease of deployment and connection flexibility by enabling the establishment of software-defined overlays in combination with optional traffic-engineered paths. Distributed IP control protocols on the packet infrastructure make the network self-contained and give a fast response to failures. There is no need for multiple control planes as extensions in the routing protocols make it possible to distribute RAN-specific forwarding information (including traffic engineering).

The overlay service runs on top of the packet network to create virtual network connectivity. e.g., via L2 EVPN or L3 IP-VPN services. The MP-BGP protocol and an SDN controller may drive the configuration and control of virtual networks and services. They can fully control traffic steering and traffic engineering for the virtual service by utilizing, for example, segment routing traffic engineering (SR-TE) capabilities of the transport infrastructure.

The distribution of functions to different sites for tailored services drives the need to simplify the deployment and operation of both RAN and its related transport. A tight interaction between the two is necessary to achieve cost-efficient deployment and operation.

Another essential new requirement is to provide a synchronization service in the new architecture. When transitioning from CPRI to eCPRI in the fronthaul, the time distribution between the baseband and O-RU has shifted from being carried by the CPRI frame itself to being carried by protocols like the packet-based precision time protocol (PTP) or Synchronous Ethernet (SyncE). ITU-T has specified several PTP profiles with relevant properties for telecom. Requirements for time alignment and frequency stability in the fronthaul network remain the same as those for CPRI-based solutions.

For the RAN to support full functionality, a performant synchronization solution must be available. With packet switches in the fronthaul network, special attention must be given to maintaining synchronization to or between radios and ensuring

sufficient synchronization availability. The transport nodes can distribute sync and hold-over functionality located in the RAN nodes (baseband, radio) or outside (packet nodes). to mitigate sync interruptions from e.g., GNSS-derived sources.

The packet fronthaul transport is a cornerstone for future automation, because of (1) simplified deployment and maintenance via SDN orchestration, (2) service-aware routing and traffic engineering, and (3) compatibility with cloud-native RAN and containerized vRAN functions.

Packet transport is not just a replacement for CPRI, but a whole transport network transformation:

- Supports both DRAN and CRAN deployments.
- Optimizes OPEX and CAPEX by enabling multiplexing and centralization.
- Provides a scalable foundation for 6G-ready architectures with integrated synchronization, SDN, and cloud-native capabilities.

The scope of the Technical Specification includes:

- To provide functional reference architecture and nodal requirements for realizing transport services for the O-RAN fronthaul interfaces in the converged packet-switching transport networks. The packet-switching transport consists of one or more packet transport functions between the fronthaul radio and baseband equipment, e.g., O-RU and O-DU. Applicability to the fronthaul interfaces other than those defined in O-RAN is not precluded.
- To define end-to-end reference architecture for transport network solution based on a packet-switched technology and services addressing control, user, and management traffic in support of the packet-switching fronthaul transport.
- Transport for RAN synchronization distribution in packet-switching fronthaul.
- Security in fronthaul, including all planes - data, control, management, and synchronization.
- Produce industry awareness and liaisons to relevant SDOs.

2 References

2.1 Normative references

- | | |
|-----|--|
| [1] | O-RAN.WG1.TS.OAD: "O-RAN Architecture Description". |
| [2] | O-RAN.WG1.TS.Slicing-Architecture-R004-v13.01: "O-RAN Slicing Architecture 13.01". |
| [3] | 3GPP, 3GPP TS 38.401, NG-RAN; "Architecture description". |
| [4] | Common Public Radio Interface V2.0, 2019: "eCPRI Interface Specification". |
| [5] | O-RAN.WG4.TS.CUS.0-R004-v18.00: "Control, User and Synchronization Plane Specification". |

2.2 Informative references

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

- | | |
|-------|---|
| [i1] | 3GPP TR 21.905: "Vocabulary for 3GPP Specifications". |
| [i2] | ETSI 20180426 Version 1.1: "ETSI PAS Process Guide". |
| [i3] | IEEE: "2021 IEEE SA Standards Style Manual". |
| [i4] | ITU-T A-1500: "Author's guide for drafting ITU-T Recommendations". |
| [i5] | ITU-T 14 February 2020: "Manual for rapporteurs and editors". |
| [i6] | IETF RFC 2119: "Key words for use in RFCs to Indicate Requirement Levels". |
| [i7] | IETF RFC 8174: "Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words". |
| [i8] | IEEE 802.1Q-2018: "IEEE Standard for Local and metropolitan area networks--Media Access Control (MAC) Bridges and Virtual Bridged Local Area Networks". |
| [i9] | IEEE 802.3-2022: "IEEE Standard Ethernet". |
| [i10] | ITU-T G.8013/Y.1731 June 2023: "OAM functions and mechanisms for Ethernet based networks". |

- [i11] IETF RFC 5036 October 2007: “LDP Specification”.
- [i12] IETF RFC 3209 December 2001: “RSVP-TE: Extensions to RSVP for LSP Tunnels”.
- [i13] IETF RFC 8660 December 2019: “Segment Routing with the MPLS Data Plane”.
- [i14] IETF RFC 1195 December 1990: “Use of OSI IS-IS for Routing in TCP/IP and Dual Environments”.
- [i15] IETF RFC 4271 January 2006: “A Border Gateway Protocol (BGP-4)”.
- [i16] IETF RFC 7432 February 2015: “BGP MPLS-Based Ethernet VPN”.

3 Definition of terms, symbols and abbreviations

3.1 Conventions

The Architecture and Requirements for RAN Fronthaul in Converged Packet Transport Networks document uses the conventions and style conformant to ETSI Publicly Available Specification (PAS). See ETSI PAS Process Guide [i2] for details. However, the document makes numerous references to works of other organizations that comply with their own conventions and style guidance, which may or may not align with ETSI PAS. For example:

- IEEE SA - 2021 IEEE SA Standards Style Manual [i3].
- ITU-T - Author's guide for drafting ITU-T Recommendations [i4].
- ITU-T Manual for rapporteurs and editors [i5].
- IETF - Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words [i7].
- IETF - Key words for use in RFCs to Indicate Requirement Levels [i6].

Attention and care should be taken to understand and comply with the conventions of each referenced document.

3.2 Terms

Void

3.3 Symbols

Void

3.4 Abbreviations

For the purposes of the present document, the abbreviations [given in [i.1] and the following] apply:

AC	Attachement Circuit
BGP	Border Gateway Protocol
CAPEX	Capital Expence
CPRI	Common Public Radio Interface
CRAN	Centralized RAN
DRAN	Distributed RAN
ETSI PAS	ETSI Publicly Available Specification
ETSI	European Telecommunications Institute
EVPN	Ethernet VPN
EVPN	Ethernet VPN
GNSS	Global Navigation Sattelite System
IEEE	Institute of Electrical and Electronics Engineers
IETF	Internet Engineering Task Force
IGP	Interior Gateway Protocol

IP	Internet Protocol
IS-IS	Intermediate System to Intermediate System
ITU	International Telecommunication Union
ITU-T	ITU- Telecommunication Standardization Sector
L2	Layer 2
L3	Layer 3
LDP	Label Distribution Protocol
LLS	Lower Layer Split
LSP	Label Switched Path
MIMO	Multiple Input Multiple Output
MP-BGP	Multi-protocol BGP
MPLS	Multiprotocol Label Switching
O-DU	O-RAN Distributed Unit
OPEX	Operational Expenditure
O-RU	O-RAN Radio Unit
PRACH	Physical Random Access Channel
PTN	Packet Transport Network
PTP	Precision Time Protocol
RAN	Radio Access Network
SDN	Software-Defined Networking
SDO	Standard Defining Organization
SR-TE	Segment Routing Traffic Engineering
SyncE	Synchronous Ethernet
TNE	Transport Network Element
VPN	Virtual Private Network
VPN	Virtual Private Network

4 Functional planes and data flows in the RAN fronthaul

In the O-RAN architecture, a single split point, known as "7-2x", is chosen as the functional split for defining the fronthaul interface between O-DU and O-RU. The split 7-2x allows a variation in placing the precoding function either in the O-DU or in the O-RU. In other architectures, the split can be at different points. The fronthaul interface facilitates the transmission of control, user, and management data between O-DU and O-RU.

The fronthaul is divided into Control, User, and Synchronization Plane (CUS-Plane) and the Management Plane (M-Plane) to support different types of data flows. The CUS-Plane carries both real-time and non-real-time traffic, including user data and control information required for radio signal processing, while the M-Plane handles configuration, monitoring, and management features.

4.1 Control Plane (C-Plane)

Control Plane (C-Plane) handles real-time control and signaling information between O-DU and O-RU. C-Plane provides instructions on scheduling, beamforming, and resource allocation to coordinate the transmission of user data. The C-Plane also ensures that O-RU processes and transmits data according to the O-DU's instructions while supporting advanced radio features. C-Plane messages have lower bandwidth requirements, whereas they have higher reliability requirements.

The C-Plane supports the eCPRI protocol over Ethernet networks illustrated in Figure 2, which provides encapsulation and prioritization for control messages distinct from user data. C-Plane messages are not concatenated with User Plane (U-Plane) messages within the same Ethernet frame. The C-Plane protocol stack may also include IP/UDP for data transmission, if supported by both the transmitting and receiving nodes.

The C-Plane features include:

- C-Plane provides low-latency and high-reliability signaling, which is necessary for 5G features such as massive MIMO (Multiple-Input Multiple-Output) and network slicing.

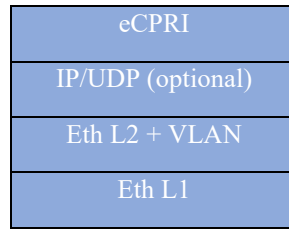


Figure 2 C-Plane Protocol Structure

- It facilitates dynamic scheduling and flexible resource allocation for radio transmissions.
- It enables real-time beamforming control for advanced antenna systems.
- With support for slot-level signaling instructions, C-Plane enables data-flow coordination and efficient and adaptive radio access.

4.2 User Plane (U-Plane)

User Plane (U-Plane) supports transport of user data i.e., IQ (In-phase Quadrature) samples representing digitized radio signals between O-DU and O-RU for uplink and downlink transmissions. U-Plane is also responsible for transferring PRACH (Physical Random Access Channel) IQ from O-RU to O-DU needed for initial UE access. U-Plane may utilize compression for IQ sample transmission to maximize throughput and efficiency of fronthaul.

The U-Plane supports the eCPRI protocol over Ethernet networks illustrated in Figure 3. The U-Plane protocol stack may also include IP/UDP for data transmission, if supported by both the transmitting and receiving nodes.

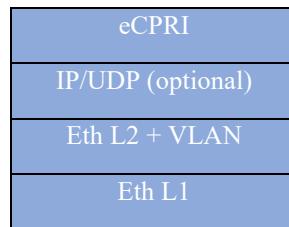


Figure 3 U-Plane Protocol Structure

The U-Plane features include:

- U-Plane provides high-bandwidth and low-latency data transfer for user data.
- It can utilize efficient compression techniques for IQ samples transmission to maximize fronthaul efficiency.
- It supports fragmentation and reassembly of O-RU to O-DU user data packets.
- U-Plane supports massive MIMO and dynamic beamforming.
- It works closely with the C-Plane to ensure data delivery aligns with dynamic scheduling and radio resource allocations.

4.3 Synchronization Plane (S-Plane)

Synchronization Plane (S-Plane) is responsible for maintaining synchronization between O-RAN components and with references such as GNSS in terms of frequency, time and phase. S-Plane achieves synchronization by distributing precise clock signals from grandmaster, PTP messages and SyncE signals as required by different synchronization configurations.

The S-Plane supports PTP and SyncE protocols over Ethernet networks illustrated in Figure 4.

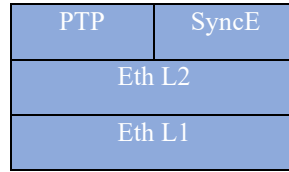


Figure 4 S-Plane Protocol Structure

The S-Plane features include:

- It maintains precise frequency, time and phase synchronization across O-RUs and O-DUs.
- S-plane provides delay compensation for fronthaul links with variable latency.
- It supports standard synchronization protocols (PTP and SyncE) for interoperability and easier integration.
- It provides precise timing alignment for advanced radio features such as TDD, carrier aggregation, MIMO and beamforming.

4.4 Management Plane (M-Plane)

Management Plane (M-Plane) responsible for non-real-time management operations of O-RU. It facilitates initialization, configuration and management of O-RU. It supports both hierarchical model (O-RU managed by O-DU) and hybrid model (O-RU managed by network management systems). To support different functionalities, various types of messages are carried over M-Plane which includes configuration management commands, performance management reports, fault management notification, and messages for software management.

M-Plane uses NETCONF/YANG as the network element management protocol and data modelling language. It requires an end-to-end IP connectivity between O-RU and the elements managing it. The M-Plane protocols stack is displayed in Figure 5.

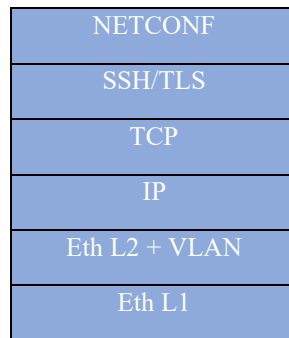


Figure 5 M-Plane Protocol Structure

The M-Plane features include:

- It enables multi-vendor support with standardized models for seamless integration.
- It supports both hierarchical and hybrid management architectures, enabling flexible deployment.
- It provides mechanisms for software upgrades, logging, and advanced diagnostics to maintain a performant and reliable RAN system.

5 Functional Reference Architecture

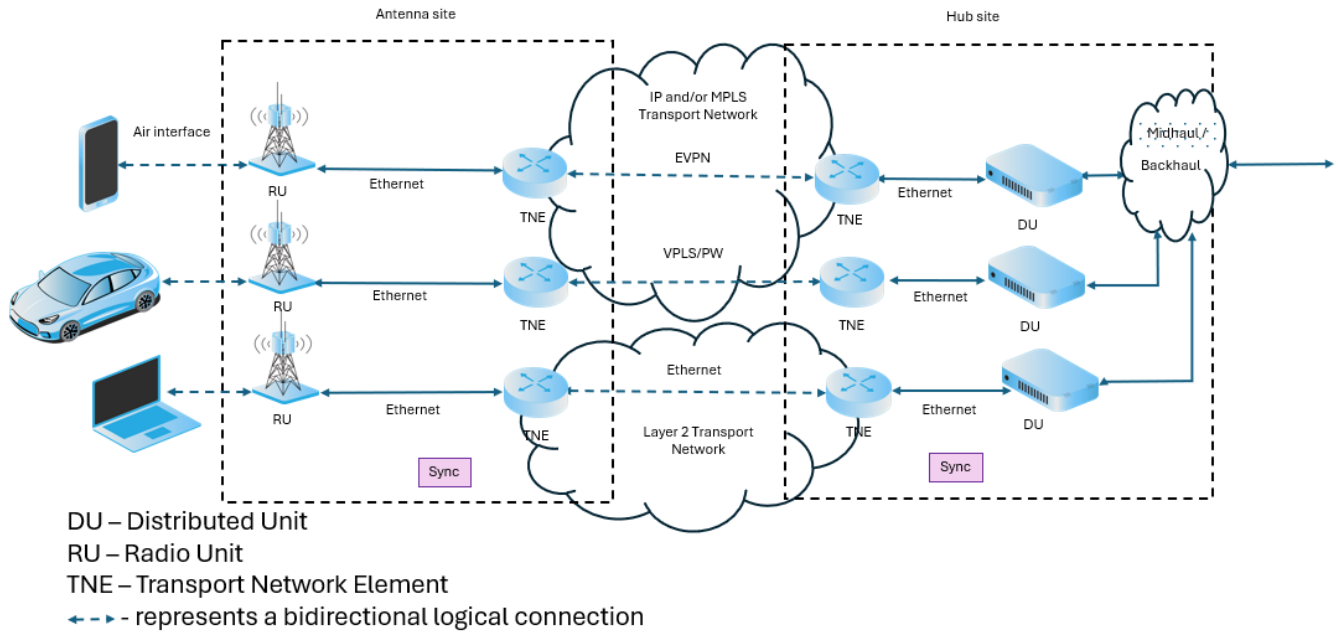


Figure 6 Reference Functional Architecture for RAN Fronthaul in Converged Packet Transport Network

A functional reference architecture is a blueprint that outlines a system's functions and their interactions, providing a generalized, reusable model that can be adapted to create specific solutions. It focuses on what the system does (the functions) without delving into the specifics of how it's implemented (the physical architecture), making it useful for design, testing, and development, as well as ensuring consistency across different projects or systems.

This project approaches the RAN Fronthaul transport architecture from the perspective of the Provider Transport Network (PTN). Mobile traffic in all planes, as noted in Section 4, is considered application (overlay) data of the respective PTN tunnels and is transparent to the PTN. In the scope of this document, transporting over the PTN the Ethernet-encapsulated traffic of the Control, User, Synchronization planes. As a result, the PTN provides transparent bridging of Ethernet frames. For this document, IP forwarding based on information in a packet that belongs to CUS-Planes of the RAN Fronthaul is outside the scope of this document.

The PTN, shown in Figure 6, provides fronthaul interface service using a combination of underlying network technologies, such as Ethernet, IP, and MPLS. Some underlay technologies, e.g., Segment Routing, allow for path engineering, thus improving utilization of network capacity. Other technologies, such as RFC 9543 Network Slicing, support resource reservation in combination with path engineering, thereby realizing the traffic engineering paradigm. In some deployment scenarios, PTN can use EVPN over Segment Routing in IPv6 (SRv6), in MPLS (SR-MPLS), or IP/MPLS underlay. In other deployments, PTN can use an Ethernet underlay network.

5.1 Provider Transport Network's demarcation

The Ethernet Attachment Circuits (AC) provide reference points for network demarcation. TNE's interfaces facing RU or DU support Ethernet AC in port and VLAN modes. In the port mode, all traffic is processed regardless of any VLAN tags present. In the latter mode, individual VLANs are treated as separate connections. That can be used with simple or multiple tags (IEEE 802.1Q [i8][i8]). Furthermore, the TNE's set of functions includes the Ethernet OAM, as defined in Link OAM (Clause 57/IEEE 802.3 [i9]) and/or ITU-T G.8013/Y.1731 [i10] [i10], that provides essential for operating the fronthaul interface at the required quality level.

5.2 Transport Network Entity functional set in the Provider Transport Network

A TNE, which is a part of the PTN that provides transport service for the fronthaul interface, includes functions in the data, control, and management planes. For example, a TNE that is part of the MPLS-based PTN supports a label distribution mechanism. Label distribution can be provided by LDP [i11], RSVP-TE [i12] for IP/MPLS networks, or by the Segment Routing extension of Interior Gateway Protocols (IGPs) for SR-MPLS [i13][i13] networks to set up LSPs. Also, labels can be provisioned over the management plane. A TNE that is a part of an IP-based PTN includes a mechanism that provides dynamic route information distribution, for example, by IS-IS [i14], as an example of IGP. In some environments, routing information can be provisioned using the paradigm of the Software Defined Network using the management plane. Segment Routing, through IGP extensions, enables path engineering for PTN. BGP [i15] can be used as the control plane of the Ethernet VPN (EVPN) [i16], a technology that provides virtual Layer 2 connectivity between dispersed sites over an IP or MPLS network.

7 Conclusions

The specification positions packet-based fronthaul not merely as a CPRI replacement but as a transformational enabler for future RAN architectures, including 6G, with integrated synchronization, SDN orchestration, and cloud-native support.

Annex (informative): Change history/Change request (history)

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Date	Revision	Description
2019.01.22	00.00.01	Create initial O-RAN version, with change-marks from previous content.
2019.01.24	00.00.02	Implemented CRs ERI-0011, NOK-0032 and DCM-0008
2019.01.31	01.00	Final version 01.00